

Nickel-Catalyzed Enantioselective Reductive Cross-Coupling Reactions

Kelsey E. Poremba, Sara E. Dibrell, and Sarah E. Reisman*



Cite This: *ACS Catal.* 2020, 10, 8237–8246



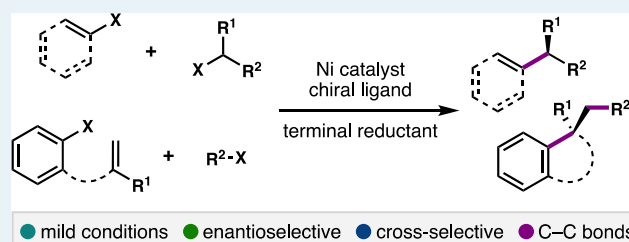
Read Online

ACCESS |

Metrics & More

Article Recommendations

ABSTRACT: Nickel-catalyzed reductive cross-coupling reactions have emerged as powerful methods to join two electrophiles. These reactions have proven particularly useful for the coupling of *sec*-alkyl electrophiles to form stereogenic centers; however, the development of enantioselective variants remains challenging. In this Perspective, we summarize the progress that has been made toward Ni-catalyzed enantioselective reductive cross-coupling reactions.



KEYWORDS: nickel catalysis, enantioselective, reductive cross-coupling, asymmetric catalysis, cross-electrophile coupling

I. INTRODUCTION

Transition-metal catalysis has unlocked new modes of reactivity that have redefined the synthetic strategies used for the preparation of enantioenriched molecules. Cross-couplings constitute one subset of transition-metal-catalyzed reactions and canonically refer to the coupling of an organic electrophile (typically an organic halide or pseudohalide) with an organometallic reagent. The use of C(sp³) coupling partners has traditionally been limited by slow oxidative addition or transmetalation, as well as decomposition via rapid β -hydride elimination in the presence of palladium or other precious metals.¹ Employing base-metal catalysts, such as nickel, for *sec*-alkyl cross-couplings can circumvent these challenges.²

Recently, Ni-catalyzed reductive cross-coupling (RCC) reactions, which join two electrophiles in the presence of a terminal reductant, have emerged as promising methods for the enantioselective coupling of C(sp³) electrophiles.³ RCC reactions typically proceed under less basic conditions at ambient temperatures (between 0 and 40 °C), which allows broad functional group tolerance and avoids the racemization of newly formed stereocenters. Given that halide electrophiles are often used as precursors to the organometallic coupling partners for canonical cross-coupling reactions, and the wide commercial availability of the halogenated building blocks, the direct use of these electrophiles in RCCs is appealing.⁴ RCC reactions can be particularly advantageous for intramolecular C–C bond formation, because they obviate the need to install both an electrophile and an organometallic functional group in the same starting material.⁵

Several challenges exist that hinder the development of reductive cross-coupling reactions. Most methods require a stoichiometric amount of heterogeneous metal dust as a terminal reductant, which renders them sensitive to stir rates,

in addition to metal purity and mesh size.^{6,7a} The generation of metal salt byproducts, as well as the common use of amide solvents, reduces the sustainability of RCCs and can introduce reproducibility issues.^{8,9} Although RCCs are widely used by medicinal chemists, advances in reductant and solvent choices will be required for application of this technology in process chemistry.^{8,10,11}

In this Perspective, we discuss the development of enantioselective RCCs catalyzed by nickel that employ a terminal reducing agent. Related reactions that are stereospecific,^{3e} that utilize photoredox cocatalysis,^{12,13} or that involve 1,2-addition to polar π systems (e.g., the Nozaki–Hiyama–Kishi coupling)¹⁴ have been reviewed elsewhere.

II. HISTORICAL CONTEXT FOR RCC REACTIONS

Seminal reports by Semmelhack,¹⁵ Kende,¹⁶ and Kumada¹⁷ demonstrated the ability of nickel to mediate the reductive homocoupling of C(sp²) halide electrophiles to form biaryl products (Scheme 1).¹⁸ However, extension of this reactivity from homocoupling to the cross-coupling of distinct partners remained elusive for several decades, due to the challenges associated with achieving cross-selectivity.¹⁹ When two electrophilic coupling partners are employed, a large excess of the less-reactive electrophile can be one way to outcompete the homocoupling process. A more efficient strategy is to

Received: April 24, 2020

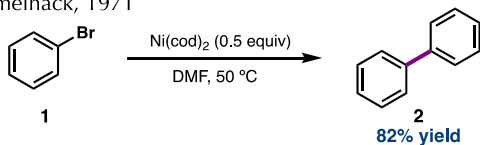
Revised: June 19, 2020

Published: June 24, 2020

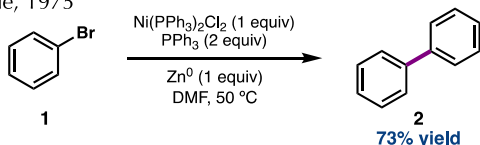


Scheme 1. Seminal Reports of Ni-Mediated Reductive Homocoupling

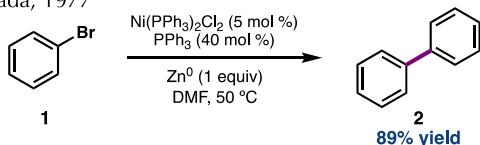
a) Semmelhack, 1971



b) Kende, 1975



c) Kumada, 1977

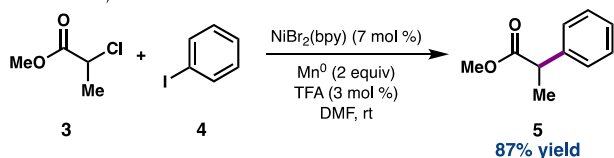


sequence the reactions of the two electrophiles, such as by leveraging the different rates of oxidative addition of a $\text{C}(\text{sp}^2)$ or $\text{C}(\text{sp}^3)$ electrophile to different Ni species in the catalytic cycle.^{20,21} If the two electrophiles react selectively with distinct oxidation states of the Ni catalyst, then sequential oxidative addition events can afford the desired cross-coupled product and minimize homocoupled dimers.²² Thus, optimization campaigns for these reactions often focus on how reaction parameters affect the distribution of the desired cross-coupled product to homodimers and reduction products.

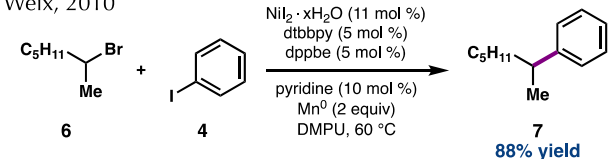
Much effort has been focused on the Ni-catalyzed cross-selective couplings of *sec*-alkyl electrophiles. In 2007, Durandetti and co-workers reported the Ni-catalyzed reductive $\text{C}(\text{sp}^2)$ – $\text{C}(\text{sp}^3)$ cross-coupling of α -chloro esters and aryl iodides using Mn^0 as a terminal reductant (Scheme 2a).²³ Weix

Scheme 2. First Reports of Ni-Catalyzed Cross-Coupling of $\text{C}(\text{sp}^2)$ and $\text{C}(\text{sp}^3)$ Electrophiles with Metal Reductants

a) Durandetti, 2007



b) Weix, 2010



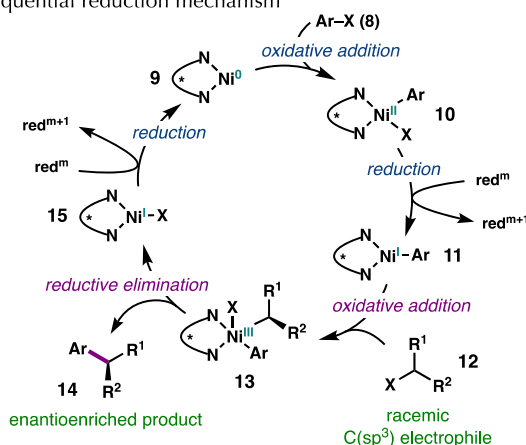
and co-workers followed in 2010 with the RCC of a *sec*-alkyl bromide and an aryl iodide, also utilizing a Ni(II) catalyst and bipyridine-based ligand (Scheme 2b).²⁴ Over the past decade, ongoing research has greatly expanded the scope of RCC reactions that use Mn^0 or Zn^0 as the terminal reductant to include many different *sec*-alkyl electrophiles, including those generated in situ from olefins.^{25,26}

III. MECHANISTIC CONSIDERATIONS

Before the past decade, all examples of Ni-catalyzed asymmetric cross-couplings fell into the category of redox-neutral transformations. Extensive method development and mechanistic investigations by Fu and co-workers on the enantioconvergent cross-coupling of *sec*-alkyl electrophiles have demonstrated the feasibility of generating an alkyl radical through halide abstraction by a Ni(I) complex and engaging this species in enantioselective catalysis.^{27,28} Our group and others hypothesized that mechanistic similarities with enantioconvergent redox-neutral couplings could be leveraged toward the development of enantioselective RCC reactions.

Investigations of Ni-catalyzed reductive cross-couplings have been conducted by several groups and can be organized into two limiting possibilities that are referred to as (1) the sequential reduction mechanism and (2) the radical chain mechanism (Figure 1).^{29,30} In a sequential reduction

a) Sequential reduction mechanism



b) Radical chain mechanism

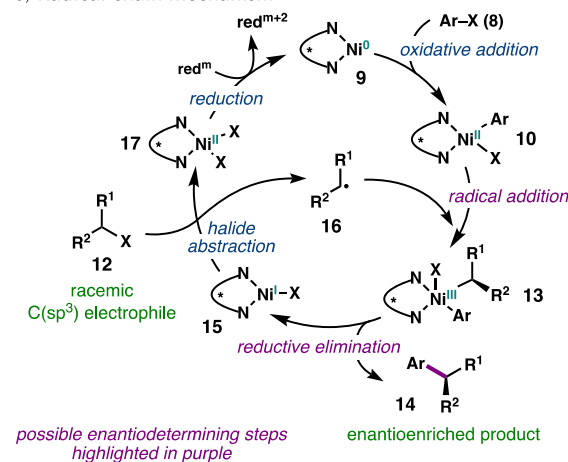


Figure 1. Proposed mechanistic hypotheses.

mechanism, it is proposed that the $\text{C}(\text{sp}^2)$ electrophile (shown as aryl halide 8 for clarity) undergoes oxidative addition to the Ni(0) species 9 to afford Ni(II)–aryl complex 10,³¹ which is then reduced by a metal reductant to 11 (Figure 1a).^{32,33} The Ni(I)–aryl complex 11 can then effect halide abstraction from the racemic *sec*-alkyl electrophile 12³⁴ to generate a prochiral radical that undergoes recombination with the metal center to give the Ni(III) intermediate 13.³⁵ Subsequent reductive elimination affords the enantioenriched

product **14** and Ni(I)–halide complex **15**, which can be reduced to regenerate the Ni(0) catalyst (**9**) and close the catalytic cycle.

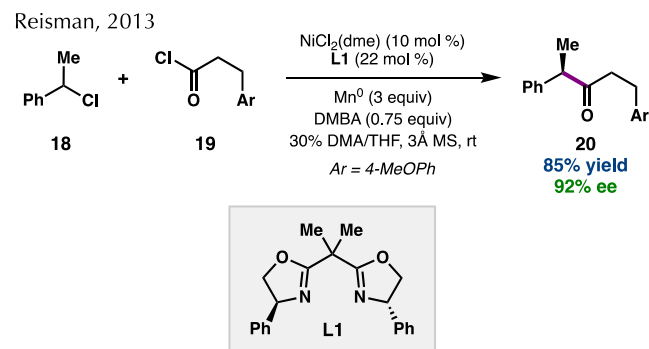
The second proposed mechanism involves a radical chain process (Figure 1b).³⁶ The C(sp²) electrophile (**8**) undergoes oxidative addition to Ni(0) complex **9**. The resulting Ni(II) intermediate **10** then combines with the cage-escaped *sec*-alkyl radical **16** to give Ni(III) complex **13**,³⁷ which upon reductive elimination gives the enantioenriched product **14** and Ni(I)–halide **15**.²⁷ The resulting Ni(I)–halide species **15** can abstract a halide from the C(sp³) electrophile **12** to generate the long-lived *sec*-alkyl radical **16**.³⁸ Finally, the Ni(II)–dihalide species **17** can be reduced, regenerating the Ni(0) catalyst **9** to close the catalytic cycle.

A major difference between the sequential reduction and radical chain mechanisms is the lifetime of the alkyl radical generated by halide abstraction, which either reacts via a radical rebound process in the solvent cage (sequential reduction mechanism) or is long-lived and escapes the cage (radical chain reaction mechanism). Experimental and computational data support each mechanism in different systems, suggesting that the mechanism of Ni-catalyzed reductive cross-couplings varies with different substrates, ligands, and reaction conditions.²¹ It is also possible that similar mechanisms are operative where the C(sp²) electrophile oxidatively adds to a Ni(I) complex, and the cycle does not proceed through reduction of the catalyst to Ni(0).^{7,26d,39} In any of these scenarios, the enantiodetermining step could be radical addition to a Ni(II) complex to form a single diastereomer of a Ni(III) complex, followed by facile reductive elimination.^{28b} Alternatively, if radical addition to Ni(II) is reversible, then reductive elimination from the Ni(III) species could be the enantiodetermining step.^{13a,35}

IV. Ni-CATALYZED ENANTIOCONVERGENT RCC REACTIONS OF C(sp²) AND C(sp³) ELECTROPHILES

In 2013, our laboratory reported the first highly enantioselective Ni-catalyzed reductive cross-coupling (Scheme 3).⁴⁰ In

Scheme 3. First Report of Enantioconvergent RCC

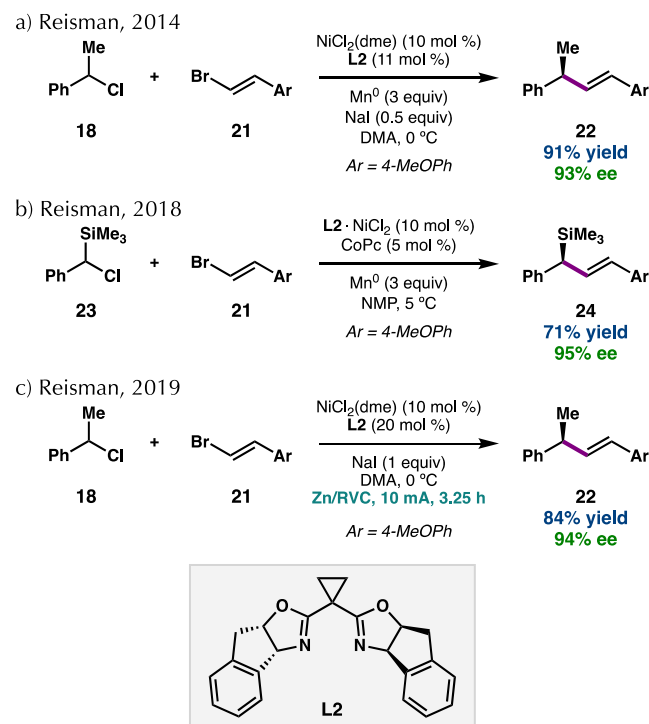


this reaction, racemic benzylic chlorides were cross-coupled with acyl chlorides using a Ni(II) precatalyst, a chiral bis(oxazoline) (BOX) ligand (**L1**), and Mn⁰ as the terminal reductant. High enantioselectivity but low reactivity was observed in THF, whereas DMA provided higher reactivity but also more homocoupling side-product formation. A mixed solvent system of DMA and THF provided the optimal balance of reactivity and selectivity. Importantly, we found that the

addition of dimethylbenzoic acid (DMBA) suppressed homocoupling of the C(sp³) electrophile. A variety of functional groups were tolerated on both coupling partners, providing the products in high yield and enantiomeric excess (ee).

In 2014, we reported a related reaction in which alkenyl bromides undergo Ni-catalyzed enantioselective RCC with benzylic chlorides (Scheme 4a).⁴¹ Chiral BOX **L2** was

Scheme 4. Enantioconvergent RCCs of Alkenyl Bromides

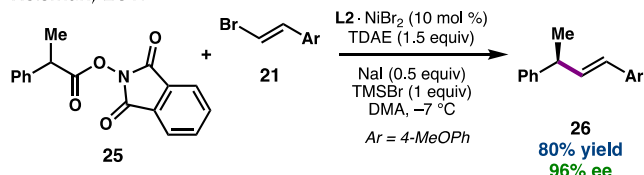


identified as the optimal ligand for this reaction, giving products bearing allylic stereocenters in excellent ee when the reaction was conducted in DMA. NaI was determined to be an important additive in the reaction, improving the yield of **22** and decreasing the formation of the dibenzyl homodimer. NaI has been suggested to enhance reactivity in reductive cross-couplings through acceleration of electron transfer between Mn⁰ and Ni or by in situ formation of iodide electrophiles.⁴² In 2018, this mode of reactivity was extended to chloro-(arylmethyl)silanes, allowing access to enantioenriched allylic silanes (Scheme 4b).⁴³ Cocatalysis with cobalt phthalocyanine (CoPc) was required for efficient coupling of these bulky silyl electrophiles, presumably to facilitate radical generation.⁴⁴

While attempts to render reductive couplings more sustainable and scalable have been reported for racemic coupling reactions, comparable asymmetric efforts are few in number.^{8,10,11} We have demonstrated that Ni-catalyzed enantioselective reductive alkenylation reactions, such as that between **18** and **21** to give **22**, can be driven electrochemically (Scheme 4c).⁴⁵ In addition, for the Ni-catalyzed asymmetric reductive alkenylation of *N*-hydroxyphthalimide (NHP) esters,^{46,47} the best results were obtained with the organic reductant tetrakis(dimethylamino)ethylene (TDAE);^{8,48} Mn⁰ and Zn⁰ as the terminal reductants provided significantly lower yields (Scheme 5).⁴⁹ The coupling of NHP esters was advantageous for improving the scope of electron-rich benzylic

Scheme 5. Enantioconvergent Reductive Decarboxylative Cross-Coupling

Reisman, 2017

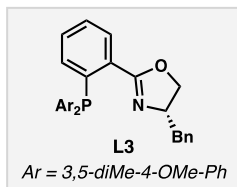
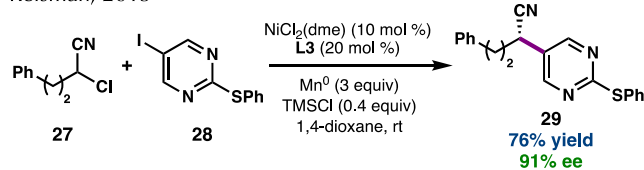


systems, where the corresponding benzylic chlorides were unstable. In the NHP ester couplings, a significant amount of (*E*)-1-(2-chlorovinyl)-4-methoxybenzene was observed when a chloride-containing precatalyst or TMSCl was used as an additive, presumably due to a Ni-catalyzed halide exchange process.⁵⁰ This alkenyl chloride was inert in the cross-coupling reaction; thus, it was necessary to eliminate all sources of chloride in the catalyst and additives to improve the yield.

Despite early success with activated $\text{C}(\text{sp}^3)$ coupling partners, variation of the $\text{C}(\text{sp}^2)$ electrophile necessitated chiral ligands outside of the BOX family. In 2015, we reported a Ni-catalyzed asymmetric RCC of α -chloronitriles and (hetero)aryl iodides (Scheme 6).⁵¹ This reaction required a

Scheme 6. Enantioconvergent RCC of α -Chloronitriles

Reisman, 2015

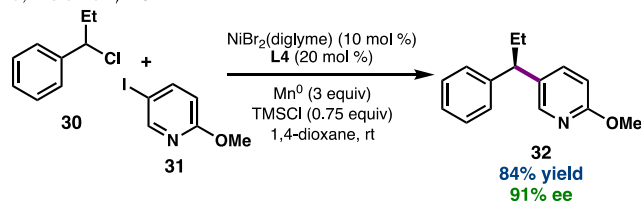


phosphino-oxazoline (PHOX) ligand (**L3**) and provided high yields and enantioselectivities of the secondary nitrile products when TMSCl was used as an additive.^{38,52} In the case of diarylalkane formation, the development of a new bioxazoline (BiOX) ligand bearing secondary alkyl substituents with long alkyl chains (**L4**) was required to obtain good yield and enantioselectivity (Scheme 7a).⁵³ Interestingly, the coupling of either α -chloronitriles or benzylic chlorides with (hetero)aryl iodides worked optimally under similar reaction conditions but required a different ligand. This highlights the importance of tuning the ligand properties when new electrophile combinations are investigated in enantioselective RCC reactions.

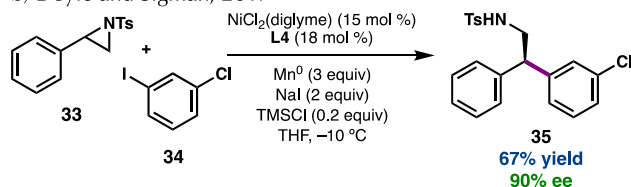
Contemporaneously with our development of the diarylalkane formation in Scheme 7a, the Doyle and Sigman groups reported an enantioselective reductive cross-coupling of racemic styrenyl-derived aziridines and aryl iodides, invoking a similar stereoconvergent mechanism (Scheme 7b).⁵⁴ Using **L4**, developed by our laboratory, 2-arylphenethylamine products were formed with high levels of enantioselectivity. Multivariate analysis of the effect of chiral BiOX ligands on the reaction revealed that ligand polarizability influences the enantioselectivity, suggesting the presence of noncovalent

Scheme 7. Enantioconvergent RCCs with a Novel BiOX Ligand

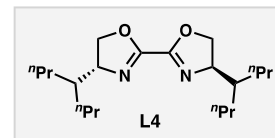
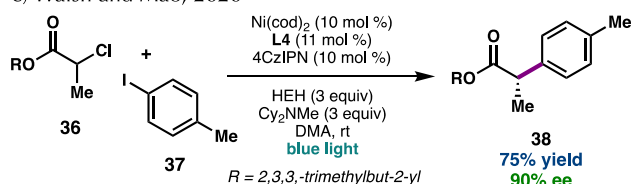
a) Reisman, 2017



b) Doyle and Sigman, 2017



c) Walsh and Mao, 2020



interactions, such as dispersion forces or $\text{CH}-\pi$ interactions, in the selectivity-determining transition state.

BiOX ligand **L4** has recently enabled the enantioconvergent RCC of α -chloro esters and aryl iodides (Scheme 7c).⁵⁵ The photoredox catalyst 1,2,3,5-tetrakis(carbazol-9-yl)-4,6-dicyanobenzene (4CzIPN) was proposed to turn over the Ni catalyst when Hantzsch ester (HEH) was employed as a soluble terminal reductant. Thus, the strategic use of photoredox cocatalysts may preclude the generation of stoichiometric metal waste by Ni-catalyzed reductive cross-couplings.

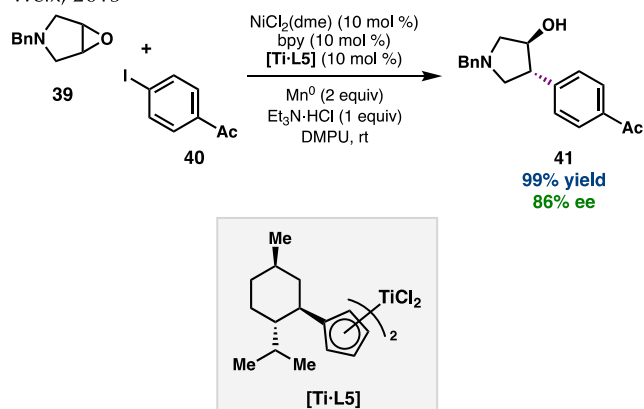
Expanding the scope of alkyl electrophiles for Ni-catalyzed asymmetric RCC reactions, the Weix group reported the enantioselective cross-coupling of *meso*-epoxides and aryl halides (Scheme 8).^{56,57} A chiral titanocene catalyst ($[\text{Ti} \cdot \text{L5}]$) was proposed to generate a β -titanoxy carbon radical from a *meso*-epoxide, which can be intercepted by a $\text{Ni}(\text{II})-\text{Ar}$ complex arising from an aryl halide. Reductive elimination from the resulting $\text{Ni}(\text{III})$ species then gives enantioenriched *trans*- β -arylcycloalkanols in excellent yields. In this transformation, the enantioselectivity is determined in the epoxide-opening step by the chiral titanocene catalyst.⁵⁶

V. Ni-CATALYZED ENANTIOSELECTIVE RCC REACTIONS OF OLEFINS

Recently, olefins have been employed in enantioselective Ni-catalyzed reductive cross-couplings to forge two C–C bonds and a stereogenic center in one reaction. These dicarbofunctionalizations are advantageous in cases where alkyl (pseudo)-halide electrophiles are unstable or require multiple steps to prepare, since the $\text{C}(\text{sp}^3)$ electrophilic fragment is generated directly from an alkene and a $\text{C}(\text{sp}^2)$ halide. Most of the

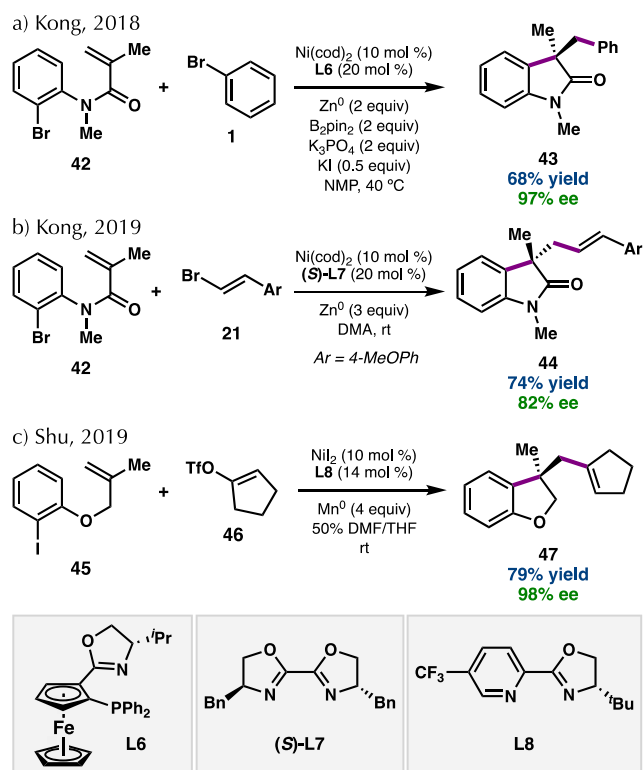
Scheme 8. Enantioconvergent RCC with Ni/Ti Cocatalysis

Weix, 2015



methods to date involve an initial intramolecular addition of a $\text{C}(\text{sp}^2)$ electrophile to an alkene. This represents a potential enantiodetermining step that distinguishes these reactions from nonconjunctive RCCs; in-depth mechanistic investigations will be instructive for future reaction development.

In 2018, Kong and co-workers disclosed the enantioselective 1,2-dicarbonylization of activated alkenes to access heterocycles bearing an all-carbon quaternary center (Scheme 9a).⁵⁸ This 1,2-diarylation required both Zn and B_2pin_2 as terminal reductants, as well as an iodide source (KI) to improve the yield. The phosphinoferrocenyloxazoline ligand **L6** induced high levels of enantioselectivity of the products, which featured various arene substitutions and tolerance of a few sterically bulky groups at the benzylic position. Similar olefin substrates were found to undergo asymmetric 1,2-

Scheme 9. Enantioselective RCCs of Olefins and $\text{C}(\text{sp}^2)$ Electrophiles

arylalkenylation with alkenyl bromide coupling partners (Scheme 9b).⁵⁹ In this case, chiral BiOX **L7** could be used in the absence of additives to provide oxindoles in good ee.

In 2019, Shu and co-workers published a related reductive transformation able to couple unactivated olefins with alkenyl triflates (Scheme 9c).⁶⁰ With the use of a pyridyloxazoline ligand (**PyOx**, **L8**), Mn^0 as the stoichiometric reductant, and each electrophile in an equimolar amount, this reaction gives heterocyclic products in moderate to good yields and excellent ee values. While this transformation successfully coupled a range of aryl substituents on the alkene partner, only 1,1-disubstitution of the alkene was tolerated.

Key to these processes is the ability of the catalyst to sequentially engage the olefin and cross-coupling partner. In a redox-neutral system, Fu and co-workers demonstrated that intermediate organonickel species can rapidly undergo olefin insertion to form a five-membered ring that is able to capture an electrophile in an enantioselective fashion.⁶¹ The reductive two-component couplings are thought to proceed via analogous mechanisms.^{58,60} Oxidative addition of the aryl halide (42 or 45) followed by reduction is proposed to access a $\text{Ni}(\text{I})$ -aryl species. This intermediate can undergo migratory insertion of the pendant alkene, which may be the enantiodetermining step. The $\text{Ni}(\text{I})$ -alkyl species resulting from this 5-*exo*-trig cyclization is then poised to undergo oxidative addition of the $\text{C}(\text{sp}^2)$ coupling partner (1 or 46) to furnish the final product 43 or 47, respectively, with high levels of enantioselectivity.

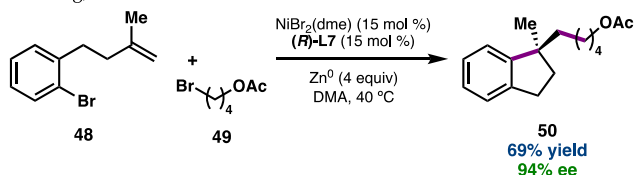
$\text{C}(\text{sp}^3)$ electrophiles have also shown competence in olefin RCCs. Wang and co-workers reported the reductive 1,2-arylalkylation and 1,2-arylbenzylation of unactivated olefins to form enantioenriched benzene-fused cyclic products (Scheme 10a,b).⁶² While chiral BiOX ligand **L7** was required for primary bromides,^{62a} the coupling of benzylic chlorides was optimal with **PyOx** **L9**.^{62b} These reactions are notable for their ability to form indane products; however, the corresponding tetralins are inaccessible, and tetrahydroisoquinolines were formed with significantly reduced ee values, indicating the difficulty of 6-*exo*-trig cyclization. These limitations highlight an opportunity for development to access products featuring other ring sizes.

Soon after, the Wang group demonstrated the ability to couple styrene-tethered acyl chlorides and $\text{C}(\text{sp}^3)$ electrophiles (Scheme 10c).⁶³ The reaction, which proceeds with Mn^0 as the terminal reductant, was found to tolerate groups of varying steric bulk at the benzylic position of 54. Competent coupling partners included primary and secondary alkyl iodides and benzyl chloride. Although the heterocyclic products were available in moderate to good yields with **PyOx** **L9**, morpholino-substituted **PyOx** **L10** was necessary to obtain good levels of enantioselectivity.

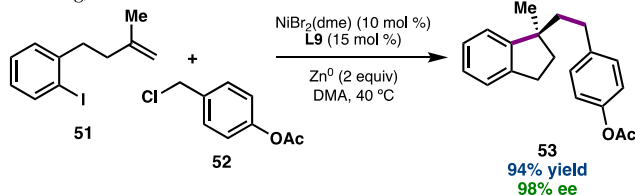
In 2019, the Diao group disclosed the first intermolecular enantioselective 1,2-dicarbonylization of activated alkenes, using BiOX **L11** (Scheme 11a).⁶⁴ Interestingly, catalytic amounts of an *N*-oxyl radical additive (**ABNO**) enabled the cross-coupling of styrenes and aryl halides to proceed with consistent and high enantioselectivities. Formation of the dibenzyl homodimer of 57 suggests the presence of an intermediate benzylic radical. In addition, stereochemical results and radical clock experiments support a mechanism involving reversible homolysis of the Ni -alkyl bond resulting from olefin migratory insertion, which may precede the enantiodetermining reductive elimination.

Scheme 10. Enantioselective RCCs of Olefins and C(sp³) Electrophiles

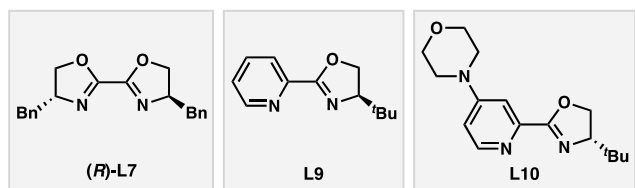
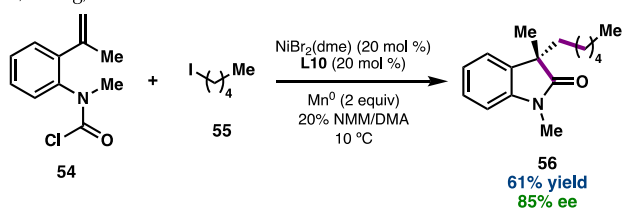
a) Wang, 2019



b) Wang, 2020

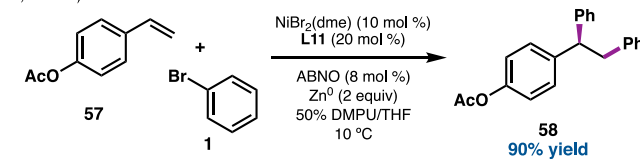


c) Wang, 2020

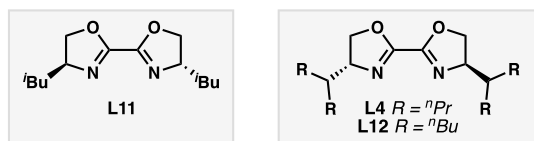
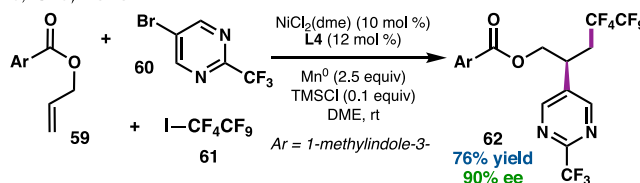


Scheme 11. Enantioselective Reductive Intermolecular Cross-Coupling of Olefins

a) Diao, 2019



b) Chu, 2020



In the following year, Chu and co-workers reported the intermolecular reductive coupling of olefins with (hetero)aryl bromides and perfluorinated alkyl iodides (Scheme 11b).⁶⁵ The use of a pendant directing group facilitated the regiospecific reaction of unactivated alkenes. Chiral BiOX ligands were found to be uniquely effective in this three-component reaction; while the previously developed L4

promoted formation of the 1,2-fluoroalkylated products in high yields, extending the alkyl chains of the ligand (L12) did not result in enhanced enantioselectivity. This transformation is an important advance from intramolecular olefin RCCs; the difunctionalization of olefins with distinct electrophiles will continue to be an interesting and significant extension of this intermolecular methodology.

VI. CONCLUDING REMARKS AND OUTLOOK

Efficient C–C bond construction through Ni-catalyzed enantioselective RCC reactions affords valuable enantioenriched small molecules from simple electrophile precursors. We anticipate that addressing several remaining challenges will be required for further advances in the field. The development of new ligand scaffolds will likely be crucial to enhancing the yield and ee of new reactions. Importantly, techniques such as ligand parametrization with multivariate linear regression analysis may draw connections between seemingly scattered data to reveal important trends in reactivity and stereoselectivity. In addition, transitioning away from heterogeneous metal reductants may increase the industrial use of reductive cross-couplings, as well as facilitate high-throughput screening for development and use of these transformations.

Activated alkyl coupling partners currently dominate the enantioselective RCCs of C(sp²) and C(sp³) electrophiles, and several limitations within this category remain. *ortho*-Substituted and *ortho,ortho*-disubstituted benzylic electrophiles exhibit low reactivity, as do those featuring sterically bulky α substituents.⁴⁰ The poor stability of electron-rich benzylic halides and α -heteroatom-substituted halides diminishes their utility.⁴ Unactivated and tertiary halides remain a significant challenge in enantioselective transformations. Thus, diversifying the pool of competent alkyl (pseudo)halide electrophiles is an important future focus.

To access a broader scope of C(sp³) coupling partners that can serve as alkyl radical precursors, radical generation mechanisms other than halogen abstraction should be explored. For example, using synergistic photoredox/Ni catalysis for C–H functionalization is an exciting new direction; however, it has been challenging to render these reactions enantioselective.^{66,67} Ultimately, the development of new methods of C(sp³) radical generation will improve the accessibility and synthetic utility of enantioselective RCCs.

Reductive olefin dicarbofunctionalization reactions offer strategic complementarity to the RCC of (pseudo)halide electrophiles. In principle, unactivated olefins can be leveraged to forge stereocenters remote from α -stabilizing groups, which would diverge from the reactivity of activated halides. An advantage of using olefin coupling partners is the ability to access all-carbon quaternary centers, which has yet to be realized in enantioconvergent RCCs. Although current methods are restricted to cyclization of five-membered rings as a strategy to effectively discriminate electrophiles, the recent development of intermolecular olefin RCCs suggests that this is not an intrinsic limitation. Further development of formally three-component couplings will rely on deeper mechanistic understanding to address challenges of electrophile differentiation.

Overall, transition-metal-catalyzed cross-coupling reactions remain an invaluable tool for the synthesis of small molecules and natural products. In particular, Ni-catalyzed reductive cross-couplings have enabled the development of mild reaction conditions that give the desired products in good yields with

high levels of enantioselectivity. We are confident that this field will continue to grow and revolutionize the way that carbon–carbon bonds are constructed in an enantioselective manner.

AUTHOR INFORMATION

Corresponding Author

Sarah E. Reisman – The Warren and Katharine Schlinger Laboratory for Chemistry and Chemical Engineering, Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, California 91125, United States; orcid.org/0000-0001-8244-9300; Email: reisman@caltech.edu

Authors

Kelsey E. Poremba – The Warren and Katharine Schlinger Laboratory for Chemistry and Chemical Engineering, Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, California 91125, United States

Sara E. Dibrell – The Warren and Katharine Schlinger Laboratory for Chemistry and Chemical Engineering, Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, California 91125, United States

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acscatal.0c01842>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

Fellowship support was provided by the National Science Foundation (graduate research fellowship to K.E.P., S.E.D. Grant No. DGE-1144469). S.E.R. is a Heritage Medical Research Institute Investigator. Financial support from the NIH (R35GM118191-01) is gratefully acknowledged.

REFERENCES

- (1) (a) Rudolph, A.; Lautens, M. Secondary Alkyl Halides in Transition-Metal-Catalyzed Cross-Coupling Reactions. *Angew. Chem., Int. Ed.* **2009**, *48* (15), 2656–2670. (b) Jana, R.; Pathak, T. P.; Sigman, M. S. Advances in Transition Metal (Pd,Ni,Fe)-Catalyzed Cross-Coupling Reactions Using Alkyl-Organometallics as Reaction Partners. *Chem. Rev.* **2011**, *111* (3), 1417–1492.
- (2) Tasker, S. Z.; Standley, E. A.; Jamison, T. F. Recent Advances in Homogeneous Nickel Catalysis. *Nature* **2014**, *509* (7500), 299–309.
- (3) For selected reviews of reductive cross-couplings, see: (a) Knappke, C. E. I.; Grupe, S.; Gärtner, D.; Corpet, M.; Gosmini, C.; Jacobi von Wangelin, A. Reductive Cross-Coupling Reactions between Two Electrophiles. *Chem. - Eur. J.* **2014**, *20* (23), 6828–6842. (b) Weix, D. J. Methods and Mechanisms for Cross-Electrophile Coupling of Csp² Halides with Alkyl Electrophiles. *Acc. Chem. Res.* **2015**, *48* (6), 1767–1775. (c) Gu, J.; Wang, X.; Xue, W.; Gong, H. Nickel-Catalyzed Reductive Coupling of Alkyl Halides with Other Electrophiles: Concept and Mechanistic Considerations. *Org. Chem. Front.* **2015**, *2*, 1411–1421. (d) Wang, X.; Dai, Y.; Gong, H. Nickel-Catalyzed Reductive Couplings. *Top. Curr. Chem.* **2016**, *374* (4), 43. (e) Lucas, E. L.; Jarvo, E. R. Stereospecific and Stereoconvergent Cross-Couplings between Alkyl Electrophiles. *Nat. Rev. Chem.* **2017**, *1* (9), No. 0065. (f) Richmond, E.; Moran, J. Recent Advances in Nickel Catalysis Enabled by Stoichiometric Metallic Reducing Agents. *Synthesis* **2018**, *50* (3), 499–513. (g) Goldfogel, M. J.; Huang, L.; Weix, D. J. Cross-Electrophile Coupling: Principles and New Reactions. In *Nickel Catalysis in Organic Synthesis*; Ogoshi, S., Ed.; Wiley: 2020; pp 183–222.
- (4) Dombrowski, A. W.; Gesmundo, N. J.; Aguirre, A. L.; Sarris, K. A.; Young, J. M.; Bogdan, A. R.; Martin, M. C.; Gedeon, S.; Wang, Y. Expanding the Medicinal Chemist Toolbox: Comparing Seven C(Sp²)-C(Sp³) Cross-Coupling Methods by Library Synthesis. *ACS Med. Chem. Lett.* **2020**, *11* (4), 597–604.
- (5) For selected examples of Ni-catalyzed intramolecular RCCs, see: (a) Yan, C.-S.; Peng, Y.; Xu, X.-B.; Wang, Y.-W. Nickel-Mediated Inter- and Intramolecular Reductive Cross-Coupling of Unactivated Alkyl Bromides and Aryl Iodides at Room Temperature. *Chem. - Eur. J.* **2012**, *18* (19), 6039–6048. (b) Xue, W.; Xu, H.; Liang, Z.; Qian, Q.; Gong, H. Nickel-Catalyzed Reductive Cyclization of Alkyl Dihalides. *Org. Lett.* **2014**, *16* (19), 4984–4987. (c) Konev, M. O.; Hanna, L. E.; Jarvo, E. R. Intra- and Intermolecular Nickel-Catalyzed Reductive Cross-Electrophile Coupling Reactions of Benzylic Esters with Aryl Halides. *Angew. Chem., Int. Ed.* **2016**, *55* (23), 6730–6733. (d) Lucas, E. L.; Hewitt, K. A.; Chen, P.-P.; Castro, A. J.; Hong, X.; Jarvo, E. R. Engaging Sulfonamides: Intramolecular Cross-Electrophile Coupling Reaction of Sulfonamides with Alkyl Chlorides. *J. Org. Chem.* **2020**, *85* (4), 1775–1793. (e) Sanford, A. B.; Thane, T. A.; McGinnis, T. M.; Chen, P.-P.; Hong, X.; Jarvo, E. R. Nickel-Catalyzed Alkyl–Alkyl Cross-Electrophile Coupling Reaction of 1,3-Dimesylates for the Synthesis of Alkylcyclopropanes. *J. Am. Chem. Soc.* **2020**, *142* (11), 5017–5023.
- (6) Yin, J.; Maguire, C. K.; Yasuda, N.; Brunskill, A. P. J.; Klapars, A. Impact of Lead Impurities in Zinc Dust on the Selective Reduction of a Dibromimidazole Derivative. *Org. Process Res. Dev.* **2017**, *21* (1), 94–97.
- (7) (a) Lin, Q.; Diao, T. Mechanism of Ni-Catalyzed Reductive 1,2-Dicarbofunctionalization of Alkenes. *J. Am. Chem. Soc.* **2019**, *141* (44), 17937–17948. (b) Diccianni, J.; Lin, Q.; Diao, T. Mechanisms of Nickel-Catalyzed Coupling Reactions and Applications in Alkene Functionalization. *Acc. Chem. Res.* **2020**, *53* (4), 906–919.
- (8) Anka-Lufford, L. L.; Huihui, K. M. M.; Gower, N. J.; Ackerman, L. K. G.; Weix, D. J. Nickel-Catalyzed Cross-Electrophile Coupling with Organic Reductants in Non-Amide Solvents. *Chem. - Eur. J.* **2016**, *22* (33), 11564–11567.
- (9) Byrne, F. P.; Jin, S.; Paggiola, G.; Petchey, T. H. M.; Clark, J. H.; Farmer, T. J.; Hunt, A. J.; Robert McElroy, C.; Sherwood, J. Tools and Techniques for Solvent Selection: Green Solvent Selection Guides. *Sustainable Chem. Processes* **2016**, *4* (1), 7.
- (10) For a recent method showcasing the utility of electrochemistry on scale, see: Perkins, R. J.; Hughes, A. J.; Weix, D. J.; Hansen, E. C. Metal-Reductant-Free Electrochemical Nickel-Catalyzed Couplings of Aryl and Alkyl Bromides in Acetonitrile. *Org. Process Res. Dev.* **2019**, *23* (8), 1746–1751.
- (11) For a recent large-scale example of heterogeneous RCC, see: Nimmagadda, S. K.; Korapati, S.; Dasgupta, D.; Malik, N. A.; Vinodini, A.; Gangu, A. S.; Kalidindi, S.; Maity, P.; Bondigela, S. S.; Venu, A.; Gallagher, W. P.; Aytar, S.; González-Bobes, F.; Vaidyanathan, R. Development and Execution of an Ni(II)-Catalyzed Reductive Cross-Coupling of Substituted 2-Chloropyridine and Ethyl 3-Chloropropanoate. *Org. Process Res. Dev.* **2020**, *24*, 1141.
- (12) (a) Milligan, J. A.; Phelan, J. P.; Badir, S. O.; Molander, G. A. Alkyl Carbon–Carbon Bond Formation by Nickel/Photoredox Cross-Coupling. *Angew. Chem., Int. Ed.* **2019**, *58* (19), 6152–6163. (b) Zhang, H.-H.; Chen, H.; Zhu, C.; Yu, S. A Review of Enantioselective Dual Transition Metal/Photoredox Catalysis. *Sci. China: Chem.* **2020**, *63*, 637–647.
- (13) For selected examples of enantioselective photoredox/Ni-catalyzed cross-couplings, see: (a) Gutierrez, O.; Tellis, J. C.; Primer, D. N.; Molander, G. A.; Kozlowski, M. C. Nickel-Catalyzed Cross-Coupling of Photoredox-Generated Radicals: Uncovering a General Manifold for Stereoconvergence in Nickel-Catalyzed Cross-Couplings. *J. Am. Chem. Soc.* **2015**, *137* (15), 4896–4899. (b) Zuo, Z.; Cong, H.; Li, W.; Choi, J.; Fu, G. C.; MacMillan, D. W. C. Enantioselective Decarboxylative Arylation of α -Amino Acids via the Merger of Photoredox and Nickel Catalysis. *J. Am. Chem. Soc.* **2016**, *138* (6), 1832–1835. (c) Stache, E. E.; Rovis, T.; Doyle, A. G. Dual Nickel- and Photoredox-Catalyzed Enantioselective Desymmetrization of Cyclic Meso-Anhydrides. *Angew. Chem., Int. Ed.* **2017**, *56* (13), 3679–3683. (d) Gandolfo, E.; Tang, X.; Roy, S. R.; Melchiorre, P.

Photochemical Asymmetric Nickel-Catalyzed Acyl Cross-Coupling. *Angew. Chem., Int. Ed.* **2019**, *58* (47), 16854–16858. (e) Pezzetta, C.; Bonifazi, D.; Davidson, R. W. M. Enantioselective Synthesis of N-Benzylic Heterocycles: A Nickel and Photoredox Dual Catalysis Approach. *Org. Lett.* **2019**, *21* (22), 8957–8961.

(14) (a) Moragas, T.; Correa, A.; Martin, R. Metal-Catalyzed Reductive Coupling Reactions of Organic Halides with Carbonyl-Type Compounds. *Chem. - Eur. J.* **2014**, *20* (27), 8242–8258. (b) Tortajada, A.; Juliá-Hernández, F.; Börjesson, M.; Moragas, T.; Martin, R. Transition-Metal-Catalyzed Carboxylation Reactions with Carbon Dioxide. *Angew. Chem., Int. Ed.* **2018**, *57* (49), 15948–15982. (c) Gil, A.; Fernando, A.; Álvarez, M. Role of the Nozaki–Hiyama–Takai–Kishi reaction in the Synthesis of Natural Products. *Chem. Rev.* **2017**, *117* (12), 8420–8446.

(15) Semmelhack, M. F.; Helquist, P. M.; Jones, L. D. Synthesis with Zerovalent Nickel. Coupling of Aryl Halides with Bis(1,5-Cyclooctadiene)Nickel(0). *J. Am. Chem. Soc.* **1971**, *93* (22), 5908–5910.

(16) Kende, A. S.; Liebeskind, L. S.; Braitsch, D. M. In Situ Generation of a Solvated Zerovalent Nickel Reagent. Biaryl Formation. *Tetrahedron Lett.* **1975**, *16* (39), 3375–3378.

(17) Zembayashi, M.; Tamao, K.; Yoshida, J.; Kumada, M. Nickel-Phosphine Complex-Catalyzed Homo Coupling of Aryl Halides in the Presence of Zinc Powder. *Tetrahedron Lett.* **1977**, *18* (47), 4089–4092.

(18) For the first reports of metal-mediated reductive homocoupling, see: (a) Wurtz, A. Sur une nouvelle classe de radicaux organiques. *Ann. Chim. Phys.* **1855**, *44*, 275–312. (b) Wurtz, A. Ueber eine neue Klasse organischer Radicale. *Ann. Chem. Pharm.* **1855**, *96* (3), 364–375.

(19) For a seminal Ni-catalyzed C(sp²)–C(sp²) cross-selective coupling with a metal reductant, see: Fürstner, A.; Shi, N. A Multicomponent Redox System Accounts for the First Nozaki–Hiyama–Kishi Reactions Catalytic in Chromium. *J. Am. Chem. Soc.* **1996**, *118* (10), 2533–2534.

(20) For the first reports of C(sp²)–C(sp³) cross-couplings with a metal reductant, see: (a) Tollens, B.; Fittig, R. Ueber die Synthese der Kohlenwasserstoffe der Benzolreihe. *Liebigs Ann.* **1864**, *131* (3), 303–323. (b) Fittig, R.; König, J. Ueber das Aethyl- und Diäthylbenzol. *Liebigs Ann.* **1867**, *144* (3), 277–294.

(21) Everson, D. A.; Weix, D. J. Cross-Electrophile Coupling: Principles of Reactivity and Selectivity. *J. Org. Chem.* **2014**, *79* (11), 4793–4798.

(22) Amatore, C.; Jutand, A.; Périchon, J.; Rollin, Y. Mechanism of the Nickel-Catalyzed Electrosynthesis of Ketones by Heterocoupling of Acyl and Benzyl Halides. *Monatsh. Chem.* **2000**, *131* (12), 1293–1304.

(23) Durandetti, M.; Gosmini, C.; Périchon, J. Ni-Catalyzed Activation of α -Chloroesters: A Simple Method for the Synthesis of α -Arylesters and β -Hydroxyesters. *Tetrahedron* **2007**, *63* (5), 1146–1153.

(24) Everson, D. A.; Shrestha, R.; Weix, D. J. Nickel-Catalyzed Reductive Cross-Coupling of Aryl Halides with Alkyl Halides. *J. Am. Chem. Soc.* **2010**, *132* (3), 920–921.

(25) For selected examples of Ni-catalyzed RCCs of C(sp²) and sec-alkyl electrophiles, see: (a) Wu, F.; Lu, W.; Qian, Q.; Ren, Q.; Gong, H. Ketone Formation via Mild Nickel-Catalyzed Reductive Coupling of Alkyl Halides with Aryl Acid Chlorides. *Org. Lett.* **2012**, *14*, 3044–3047. (b) Wang, S.; Qian, Q.; Gong, H. Nickel-Catalyzed Reductive Coupling of Aryl Halides with Secondary Alkyl Bromides and Allylic Acetate. *Org. Lett.* **2012**, *14* (13), 3352–3355. (c) Wotal, A. C.; Weix, D. J. Synthesis of Functionalized Dialkyl Ketones from Carboxylic Acid Derivatives and Alkyl Halides. *Org. Lett.* **2012**, *14* (6), 1476–1479. (d) Zhao, Y.; Weix, D. J. Nickel-Catalyzed Regiodivergent Opening of Epoxides with Aryl Halides: Co-Catalysis Controls Regioselectivity. *J. Am. Chem. Soc.* **2014**, *136* (1), 48–51. (e) Molander, G. A.; Traister, K. M.; O'Neill, B. T. Reductive Cross-Coupling of Nonaromatic, Heterocyclic Bromides with Aryl and Heteroaryl Bromides. *J. Org. Chem.* **2014**, *79* (12), 5771–5780.

(f) Arendt, K. M.; Doyle, A. G. Dialkyl Ether Formation via Nickel-Catalyzed Cross Coupling of Acetals and Aryl Iodides. *Angew. Chem., Int. Ed.* **2015**, *54*, 9876–9880. (g) Molander, G. A.; Traister, K. M.; O'Neill, B. T. Engaging Nonaromatic, Heterocyclic Tosylates in Reductive Cross-Coupling with Aryl and Heteroaryl Bromides. *J. Org. Chem.* **2015**, *80* (3), 2907–2911. (h) Qiu, C.; Yao, K.; Zhang, X.; Gong, H. Ni-Catalyzed Reductive Coupling of α -Halocarbonyl Derivatives with Vinyl Bromides. *Org. Biomol. Chem.* **2016**, *14* (48), 11332–11335. (i) Gu, J.; Qiu, C.; Lu, W.; Qian, Q.; Lin, K.; Gong, H. Nickel-Catalyzed Reductive Cross-Coupling of Vinyl Bromides with Unactivated Alkyl Halides. *Synthesis* **2017**, *49* (8), 1867–1873. (j) Gao, M.; Sun, D.; Gong, H. Ni-Catalyzed Reductive C–O Bond Arylation of Oxalates Derived from α -Hydroxy Esters with Aryl Halides. *Org. Lett.* **2019**, *21*, 1645–1648. (k) Martin-Montero, R.; Yatham, V. R.; Yin, H.; Davies, J.; Martin, R. Ni-Catalyzed Reductive Deaminative Arylation at Sp³ Carbon Centers. *Org. Lett.* **2019**, *21* (8), 2947–2951.

(26) For selected examples of Ni-catalyzed RCCs of olefins, see: (a) García-Domínguez, A.; Li, Z.; Nevado, C. Nickel-Catalyzed Reductive Dicarbofunctionalization of Alkenes. *J. Am. Chem. Soc.* **2017**, *139* (20), 6835–6838. (b) Kuang, Y.; Wang, X.; Anthony, D.; Diao, T. Ni-Catalyzed Two-Component Reductive Dicarbofunctionalization of Alkenes via Radical Cyclization. *Chem. Commun.* **2018**, *54* (20), 2558–2561. (c) Zhao, X.; Tu, H.-Y.; Guo, L.; Zhu, S.; Qing, F.-L.; Chu, L. Intermolecular Selective Carboacylation of Alkenes via Nickel-Catalyzed Reductive Radical Relay. *Nat. Commun.* **2018**, *9* (1), 1–7. (d) Shu, W.; García-Domínguez, A.; Quirós, M. T.; Mondal, R.; Cárdenas, D. J.; Nevado, C. Ni-Catalyzed Reductive Dicarbofunctionalization of Nonactivated Alkenes: Scope and Mechanistic Insights. *J. Am. Chem. Soc.* **2019**, *141* (35), 13812–13821. (e) Jin, Y.; Wang, C. Ni-catalyzed reductive arylalkylation of unactivated alkenes. *Chem. Sci.* **2019**, *10*, 1780–1785.

(27) Fu, G. C. Transition-Metal Catalysis of Nucleophilic Substitution Reactions: A Radical Alternative to SN1 and SN2 Processes. *ACS Cent. Sci.* **2017**, *3* (7), 692–700.

(28) (a) Schley, N. D.; Fu, G. C. Nickel-Catalyzed Negishi Arylations of Propargylic Bromides: A Mechanistic Investigation. *J. Am. Chem. Soc.* **2014**, *136* (47), 16588–16593. (b) Yin, H.; Fu, G. C. Mechanistic Investigation of Enantioconvergent Kumada Reactions of Racemic α -Bromoketones Catalyzed by a Nickel/Bis(Oxazoline) Complex. *J. Am. Chem. Soc.* **2019**, *141* (38), 15433–15440.

(29) Everson, D. A.; Jones, B. A.; Weix, D. J. Replacing Conventional Carbon Nucleophiles with Electrophiles: Nickel-Catalyzed Reductive Alkylation of Aryl Bromides and Chlorides. *J. Am. Chem. Soc.* **2012**, *134* (14), 6146–6159.

(30) Ren, Q.; Jiang, F.; Gong, H. DFT Study of the Single Electron Transfer Mechanisms in Ni-Catalyzed Reductive Cross-Coupling of Aryl Bromide and Alkyl Bromide. *J. Organomet. Chem.* **2014**, *770*, 130–135.

(31) Tsou, T. T.; Kochi, J. K. Mechanism of Oxidative Addition. Reaction of Nickel(0) Complexes with Aromatic Halides. *J. Am. Chem. Soc.* **1979**, *101* (21), 6319–6332.

(32) Amatore, C.; Jutand, A. Rates and Mechanism of Biphenyl Synthesis Catalyzed by Electrogenated Coordinatively Unsaturated Nickel Complexes. *Organometallics* **1988**, *7*, 2203–2214.

(33) Colon, I.; Kelsey, D. R. Coupling of Aryl Chlorides by Nickel and Reducing Metals. *J. Org. Chem.* **1986**, *51* (14), 2627–2637.

(34) (a) Anderson, T. J.; Jones, G. D.; Vici, D. A. Evidence for a Ni^I Active Species in the Catalytic Cross-Coupling of Alkyl Electrophiles. *J. Am. Chem. Soc.* **2004**, *126*, 8100–8101. (b) Jones, G. D.; Martin, J. L.; McFarland, C.; Allen, O. R.; Hall, R. E.; Haley, A. D.; Brandon, R. J.; Konovalova, T.; Desrochers, P. J.; Pulay, P.; Vici, D. A. Ligand Redox Effects in the Synthesis, Electronic Structure, and Reactivity of an Alkyl-Alkyl Cross-Coupling Catalyst. *J. Am. Chem. Soc.* **2006**, *128*, 13175–13183. (c) Diccianni, J. B.; Katigbak, J.; Hu, C.; Diao, T. Mechanistic Characterization of (Xantphos)Ni(I)-Mediated Alkyl Bromide Activation: Oxidative Addition, Electron Transfer, or Halogen-Atom Abstraction. *J. Am. Chem. Soc.* **2019**, *141* (4), 1788–1796.

- (35) Lin, X.; Sun, J.; Xi, Y.; Lin, D. How Racemic Secondary Alkyl Electrophiles Proceed to Enantioselective Products in Negishi Cross-Coupling Reactions. *Organometallics* **2011**, *30* (12), 3284–3292.
- (36) For a related proposal of a radical chain mechanism involving π -allyl–Ni complexes, see: Hegedus, L. S.; Miller, L. L. Reaction of π -Allylnickel Bromide Complexes with Organic Halides. Stereochemistry and Mechanism. *J. Am. Chem. Soc.* **1975**, *97* (2), 459–460.
- (37) Wang, X.; Ma, G.; Peng, Y.; Pitsch, C. E.; Moll, B. J.; Ly, T. D.; Wang, X.; Gong, H. Ni-Catalyzed Reductive Coupling of Electron-Rich Aryl Iodides with Tertiary Alkyl Halides. *J. Am. Chem. Soc.* **2018**, *140* (43), 14490–14497.
- (38) Biswas, S.; Weix, D. J. Mechanism and Selectivity in Nickel-Catalyzed Cross-Electrophile Coupling of Aryl Halides with Alkyl Halides. *J. Am. Chem. Soc.* **2013**, *135* (43), 16192–16197.
- (39) Tsou, T. T.; Kochi, J. K. Mechanism of Biaryl Synthesis with Nickel Complexes. *J. Am. Chem. Soc.* **1979**, *101* (25), 7547–7560.
- (40) Cherney, A. H.; Kadunce, N. T.; Reisman, S. E. Catalytic Asymmetric Reductive Acyl Cross-Coupling: Synthesis of Enantioenriched Acyclic α,α -Disubstituted Ketones. *J. Am. Chem. Soc.* **2013**, *135*, 7442–7445.
- (41) Cherney, A. H.; Reisman, S. E. Nickel-Catalyzed Asymmetric Reductive Cross-Coupling Between Vinyl and Benzyl Electrophiles. *J. Am. Chem. Soc.* **2014**, *136*, 14365–14368.
- (42) Prinsell, M. R.; Everson, D. A.; Weix, D. J. Nickel-Catalyzed, Sodium Iodide-Promoted Reductive Dimerization of Alkyl Halides, Alkyl Pseudohalides, and Allylic Acetates. *Chem. Commun.* **2010**, *46* (31), 5743–5745.
- (43) Hofstra, J. L.; Cherney, A. H.; Ordner, C. M.; Reisman, S. E. Synthesis of Enantioenriched Allylic Silanes via Nickel-Catalyzed Reductive Cross-Coupling. *J. Am. Chem. Soc.* **2018**, *140* (1), 139–142.
- (44) Ackerman, L. K. G.; Anka-Lufford, L. L.; Naodovic, M.; Weix, D. J. Cobalt Co-Catalysis for Cross-Electrophile Coupling: Diaryl-methanes from Benzyl Mesylates and Aryl Halides. *Chem. Sci.* **2015**, *6*, 1115–1119.
- (45) DeLano, T. J.; Reisman, S. E. Enantioselective Electroreductive Coupling of Alkenyl and Benzyl Halides via Nickel Catalysis. *ACS Catal.* **2019**, *9* (8), 6751–6754.
- (46) Huihui, K. M. M.; Caputo, J. A.; Melchor, Z.; Olivares, A. M.; Spiewak, A. M.; Johnson, K. A.; DiBenedetto, T. A.; Kim, S.; Ackerman, L. K. G.; Weix, D. J. Decarboxylative Cross-Electrophile Coupling of *N*-Hydroxyphthalimide Esters with Aryl Iodides. *J. Am. Chem. Soc.* **2016**, *138*, 5016–5019.
- (47) Ni, S.; Padial, N. M.; Kingston, C.; Vantourout, J. C.; Schmitt, D. C.; Edwards, J. T.; Kruszyk, M. M.; Merchant, R. R.; Mykhailiuk, P. K.; Sanchez, B. B.; Yang, S.; Perry, M. A.; Gallego, G. M.; Mousseau, J. J.; Collins, M. R.; Cherney, R. J.; Lebed, P. S.; Chen, J. S.; Qin, T.; Baran, P. S. A Radical Approach to Anionic Chemistry: Synthesis of Ketones, Alcohols, and Amines. *J. Am. Chem. Soc.* **2019**, *141* (16), 6726–6739.
- (48) Kuroboshi, M.; Tanaka, M.; Kishimoto, S.; Goto, K.; Mochizuki, M.; Tanaka, H. Tetrakis(dimethylamino)-ethylene (TDAE) as a Potent Organic Electron Source: Alkenylation of Aldehydes Using a Ni/Cr/TDAE Redox System. *Tetrahedron Lett.* **2000**, *41* (1), 81–84.
- (49) Suzuki, N.; Hofstra, J. L.; Poremba, K. E.; Reisman, S. E. Nickel-Catalyzed Enantioselective Cross-Coupling of *N*-Hydroxyphthalimide Esters with Vinyl Bromides. *Org. Lett.* **2017**, *19* (8), 2150–2153.
- (50) (a) Takagi, K.; Hayama, N.; Inokawa, S. Synthesis of Vinyl Iodides from Vinyl Bromides and Potassium Iodide by Means of Nickel Catalyst. *Chem. Lett.* **1978**, *7* (12), 1435–1436. (b) Tsou, T. T.; Kochi, J. K. Nickel Catalysis in Halogen Exchange with Aryl and Vinylic Halides. *J. Org. Chem.* **1980**, *45* (10), 1930–1937. (c) Hofstra, J. L.; Poremba, K. E.; Shimozone, A. M.; Reisman, S. E. Nickel-Catalyzed Conversion of Enol Triflates into Alkenyl Halides. *Angew. Chem., Int. Ed.* **2019**, *58* (42), 14901–14905.
- (51) Kadunce, N. T.; Reisman, S. E. Nickel-Catalyzed Asymmetric Reductive Cross-Coupling between Heteroaryl Iodides and α -Chloronitriles. *J. Am. Chem. Soc.* **2015**, *137*, 10480–10483.
- (52) (a) Takai, K.; Ueda, T.; Hayashi, T.; Moriwake, T. Activation of Manganese Metal by a Catalytic Amount of PbCl_2 and Me_3SiCl . *Tetrahedron Lett.* **1996**, *37* (39), 7049–7052. (b) Johnson, K. A.; Biswas, S.; Weix, D. J. Cross-Electrophile Coupling of Vinyl Halides with Alkyl Halides. *Chem. - Eur. J.* **2016**, *22* (22), 7399–7402.
- (53) Poremba, K. E.; Kadunce, N. T.; Suzuki, N.; Cherney, A. H.; Reisman, S. E. Nickel-Catalyzed Asymmetric Reductive Cross-Coupling to Access 1,1-Diaryllkanes. *J. Am. Chem. Soc.* **2017**, *139* (16), 5684–5687.
- (54) Woods, B. P.; Orlandi, M.; Huang, C.-Y.; Sigman, M. S.; Doyle, A. G. Nickel-Catalyzed Enantioselective Reductive Cross-Coupling of Styrenyl Aziridines. *J. Am. Chem. Soc.* **2017**, *139* (16), 5688–5691.
- (55) Guan, H.; Zhang, Q.; Walsh, P. J.; Mao, J. Nickel/Photoredox-Catalyzed Asymmetric Reductive Cross-Coupling of Racemic α -Chloro Esters with Aryl Iodides. *Angew. Chem., Int. Ed.* **2020**, *59* (13), 5172–5177.
- (56) Zhao, Y.; Weix, D. J. Enantioselective Cross-Coupling of Meso-Epoxides with Aryl Halides. *J. Am. Chem. Soc.* **2015**, *137* (9), 3237–3240.
- (57) A related reductive cross-coupling of chiral 3,4-epoxy alcohols with aryl iodides affords 1,1-diaryldiol products with enriched ee. See: Banerjee, A.; Yamamoto, H. Nickel Catalyzed Regio-, Diastereo-, and Enantioselective Cross-Coupling of 3,4-Epoxyalcohol with Aryl Iodides. *Org. Lett.* **2017**, *19* (16), 4363–4366.
- (58) Wang, K.; Ding, Z.; Zhou, Z.; Kong, W. Ni-Catalyzed Enantioselective Reductive Diarylation of Activated Alkenes by Domino Cyclization/Cross-Coupling. *J. Am. Chem. Soc.* **2018**, *140* (39), 12364–12368.
- (59) Li, Y.; Ding, Z.; Lei, A.; Kong, W. Ni-Catalyzed enantioselective reductive aryl-alkenylation of alkenes: application to the synthesis of (+)-physosvenine and (+)-physostigmine. *Org. Chem. Front.* **2019**, *6* (18), 3305–3309.
- (60) Tian, Z.-X.; Qiao, J.-B.; Xu, G.-L.; Pang, X.; Qi, L.; Ma, W.-Y.; Zhao, Z.-Z.; Duan, J.; Du, Y.-F.; Su, P.; Liu, X.-Y.; Shu, X.-Z. Highly Enantioselective Cross-Electrophile Aryl-Alkenylation of Unactivated Alkenes. *J. Am. Chem. Soc.* **2019**, *141* (18), 7637–7643.
- (61) Cong, H.; Fu, G. C. Catalytic Enantioselective Cyclization/Cross-Coupling with Alkyl Electrophiles. *J. Am. Chem. Soc.* **2014**, *136* (10), 3788–3791.
- (62) (a) Jin, Y.; Wang, C. Ni-Catalyzed Asymmetric Reductive Arylalkylation of Unactivated Alkenes. *Angew. Chem., Int. Ed.* **2019**, *58*, 6722–6726. (b) Jin, Y.; Yang, H.; Wang, C. Nickel-Catalyzed Asymmetric Reductive Arylbenzylation of Unactivated Alkenes. *Org. Lett.* **2020**, *22* (7), 2724–2729.
- (63) Lan, Y.; Wang, C. Nickel-Catalyzed Enantioselective Reductive Carbo-Acylation of Alkenes. *Comm. Chem.* **2020**, *3* (1), 1–9.
- (64) Anthony, D.; Lin, Q.; Baudet, J.; Diao, T. Nickel-Catalyzed Asymmetric Reductive Diarylation of Vinylarenes. *Angew. Chem., Int. Ed.* **2019**, *58* (10), 3198–3202.
- (65) Tu, H. Y.; Wang, F.; Hou, L.; Li, Y.; Zhu, S.; Zhao, X.; Li, H.; Qing, F. L.; Chu, L. Enantioselective Three-Component Fluoroalkylation of Unactivated Olefins Through Nickel-Catalyzed Cross-Electrophile Coupling. *J. Am. Chem. Soc.* **2020**, *142* (21), 9604–9611.
- (66) For isolated examples of enantioselective photoredox/Ni-catalyzed C–H functionalization, see: (a) Ahneman, D. T.; Doyle, A. G. C–H functionalization of amines with aryl halides by nickel-photoredox catalysis. *Chem. Sci.* **2016**, *7* (12), 7002–7006. (b) Shen, Y.; Gu, Y.; Martin, R. Sp^3 C–H Arylation and Alkylation Enabled by the Synergy of Triplet Excited Ketones and Nickel Catalysts. *J. Am. Chem. Soc.* **2018**, *140* (38), 12200–12209. (c) Rand, A. W.; Yin, H.; Xu, L.; Giacoboni, J.; Martin-Montero, R.; Romano, C.; Montgomery, J.; Martin, R. Dual Catalytic Platform for Enabling Sp^3 α C–H Arylation and Alkylation of Benzamides. *ACS Catal.* **2020**, *10* (8), 4671–4676.
- (67) For seminal examples of enantioselective photoredox/Ni-catalyzed C–H functionalization, see: (a) Cheng, X.; Lu, H.; Lu, Z.

Enantioselective Benzylic C–H Arylation via Photoredox and Nickel Dual Catalysis. *Nat. Commun.* **2019**, *10* (1), 1–7. (b) Fan, P.; Lan, Y.; Zhang, C.; Wang, C. Nickel/Photo-Cocatalyzed Asymmetric Acyl-Carbamoylation of Alkenes. *J. Am. Chem. Soc.* **2020**, *142* (5), 2180–2186.